

SARIWON, Korea, Democratic People's Republic ofWMO: **470650**Lat: **38.517N**Lon: **125.767E**Elev: **171**StdP: **14.61**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	10.6	-5.0	4.2	12.1	-1.0	5.3	14.9	26.9	20.1	23.9	20.1	5.0	320	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	88.9	78.4	86.7	76.7	84.7	75.3	80.6	86.9	79.1	84.7	77.8	82.7	5.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.4	84.6	77.4	143.5	83.0	76.3	138.0	81.6	44.4	87.2	42.9	85.5	41.5	82.7	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	19.7	16.8	3.4	92.7	5.1	2.9	-0.3	94.8	-3.3	96.5	-6.2	98.2	-9.9	100.3		
(5)			2.5	82.9	4.7	1.7	-0.9	84.1	-3.6	85.1	-6.3	86.0	-9.7	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	23.6	28.6	39.7	52.6	63.7	72.1	77.1	77.6	68.9	56.9	41.8	28.4
	DBStd	19.89	7.42	7.16	6.64	6.70	4.86	3.89	3.72	4.13	4.57	6.37	8.26	7.63
	HDD50	2746	818	600	327	49	0	0	0	0	0	18	265	669
	HDD65	5654	1283	1020	785	374	83	3	0	0	18	259	695	1134
	CDD50	3732	0	0	7	127	425	662	840	855	567	230	19	0
	CDD65	1166	0	0	0	2	43	215	375	390	135	6	0	0
	CDH74	8274	0	0	0	17	328	1271	2789	3185	656	28	0	0
	CDH80	2114	0	0	0	0	44	293	746	954	76	1	0	0
Wind	WSAvg	5.3	5.3	5.9	6.6	6.4	6.2	4.9	4.7	4.2	3.7	4.5	5.4	5.4
Precipitation	PrecAvg	35.7	0.4	0.6	0.7	1.9	3.1	3.6	10.5	8.2	3.8	1.7	1.4	0.7
	PrecMax	70.4	1.1	1.9	2.0	4.6	8.6	10.7	23.4	22.4	14.8	4.7	4.3	1.8
	PrecMin	15.7	0.0	0.0	0.0	0.0	0.3	0.9	3.0	0.9	0.5	0.1	0.2	0.1
	PrecStd	11.0	0.3	0.4	0.6	1.1	2.0	2.3	5.2	5.1	3.6	1.3	1.0	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	52.3	63.8	76.5	84.4	88.2	92.1	92.6	84.7	77.4	64.8	50.0
		MCWB	39.5	44.9	52.0	60.2	65.5	72.4	79.5	80.5	72.9	64.9	57.4	44.8
	2%	DB	40.1	46.3	58.6	72.0	80.3	85.6	88.7	89.5	82.3	73.3	60.7	45.8
		MCWB	35.9	39.7	48.4	57.8	64.4	71.0	78.3	80.0	71.2	62.4	54.0	41.9
	5%	DB	37.3	42.5	54.7	68.4	77.6	83.4	86.3	87.4	80.3	70.2	57.3	42.3
		MCWB	33.7	37.2	46.2	55.4	62.9	70.1	77.2	78.6	69.8	60.9	52.0	38.6
	10%	DB	34.4	39.6	51.0	64.8	74.6	80.9	84.1	85.1	78.1	67.3	54.2	39.3
		MCWB	31.2	35.1	43.6	53.9	61.8	69.1	76.5	77.2	68.5	59.2	49.3	36.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	45.8	53.4	64.6	69.8	75.7	82.0	83.0	76.2	67.4	59.6	46.6
		MCDB	43.2	49.9	61.0	73.2	78.2	84.4	88.0	89.4	81.2	74.6	62.8	48.4
	2%	WB	36.6	40.5	50.2	59.9	67.0	73.8	80.1	81.1	74.1	64.7	55.5	42.2
		MCDB	39.3	45.0	56.8	68.3	76.2	81.4	86.2	87.4	79.0	70.1	59.3	45.2
	5%	WB	34.0	37.8	47.3	57.4	65.1	72.4	78.9	79.9	72.3	62.6	52.6	38.9
		MCDB	36.7	41.8	53.5	65.6	74.0	79.8	84.3	85.9	77.5	68.3	56.9	41.8
	10%	WB	31.5	35.3	44.2	55.1	63.5	71.1	77.6	78.6	70.5	60.4	49.6	36.2
		MCDB	34.1	39.0	49.9	62.8	71.8	78.1	82.4	83.9	76.0	66.0	53.4	39.1
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.8	16.3	18.7	18.6	15.7	11.6	12.7	16.5	17.6	14.2	12.7
		MCDBR	16.4	18.9	22.0	23.7	23.8	20.2	15.5	15.2	18.4	20.3	17.9	15.5
		MCWBR	13.1	13.9	14.3	12.7	11.5	9.3	7.7	7.7	9.4	11.0	12.4	12.5
	5% WB	MCDBR	14.9	16.9	20.0	20.1	19.6	16.2	12.4	13.2	14.9	17.7	16.0	14.6
		MCWBR	12.7	13.5	14.4	12.5	11.2	8.7	7.4	7.6	9.1	10.8	12.6	12.5
Clear-Sky Solar Irradiance	taub	0.420	0.487	0.550	0.613	0.626	0.707	0.680	0.580	0.511	0.523	0.495	0.431	
	taud	1.929	1.814	1.718	1.638	1.658	1.564	1.669	1.882	1.980	1.843	1.850	1.934	
	Ebn at Noon	222	223	224	220	220	203	207	226	233	210	194	207	
	Edh at Noon	44	56	69	79	79	86	77	61	52	54	48	41	
All-Sky Solar Radiation	RadAvg	773	1014	1373	1602	1786	1650	1300	1398	1367	1120	750	661	
	RadStd	62	75	112	114	168	163	173	172	123	70	76	46	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+99

Nomenclature: See separate page